Automated Cricket Ground Control System

Group No: 20

230511A - Rajapaksha J.S.W.

230516T - Rajapakshe R.P.M.T.

230517X - Rajapakshe W.R.K.L.M.H.N

Introduction

Cricket stadiums require careful management of environmental conditions such as lighting, ground moisture, and weather changes to ensure optimal playing conditions and player safety. Traditionally, these operations are handled manually, which can lead to inefficiencies, delays, and inconsistent control.

With advancements in automation and instrumentation, it is possible to develop intelligent systems that monitor environmental parameters and respond automatically. This project focuses on designing a sensor-based automated system that integrates lighting control, irrigation management, and rain detection using an DAQ-based platform with LabVIEW monitoring.

Aim

To design and develop an automated environmental control system for a cricket stadium that intelligently monitors and regulates lighting, irrigation, and rain alerts using sensor-based inputs and centralized control.

Objectives

- To automate **floodlight operation** based on ambient light intensity.
- To detect **rainfall conditions** and provide real-time alerts.
- To monitor **soil moisture levels** and control irrigation automatically to maintain proper grass covering.
- To measure **temperature and humidity for evaluating playing conditions**
- To reduce manual intervention in ground maintenance.

Scope

This project focuses on designing and implementing a **prototype system** for smart cricket ground management using embedded systems and sensors.

The scope includes:

- Real-time monitoring of environmental parameters (light, rain, temperature, humidity).
- Soil condition monitoring for irrigation control.
- Automated operation of floodlights and water sprinklers.
- Alert system for weather changes.

Literature Review

Soil Moisture-Based Irrigation Systems for Maintaining Proper Grass Covering

This study evaluated timer-based, soil-moisture sensor (SMS), and evapotranspiration (ET) irrigation systems for turfgrass. The SMS2 system performed best overall, achieving both high efficiency and adequate irrigation, while SMS1 saved the most water but sometimes lacked sufficient irrigation. The ET system maintained good irrigation adequacy but used more water than necessary, making it the least efficient. Overall, soil-moisture–based systems provided significant water savings compared to traditional timer-based methods. [1]

Playing Condition Monitoring Using Temperature and Humidity Sensors

IoT weather integration is used in cricket grounds to continuously monitor environmental conditions and automatically manage playing surfaces. Sensors collect real-time data such as rainfall, humidity, temperature, light intensity, and soil moisture. This data is processed to support key decisions like match continuation, pitch preparation, and ground maintenance. [2]

Automatic Lighting Control Systems

A study in the field of Computer Technology investigates the application of intelligent lighting control systems in sports venues. The research analyzes the lighting requirements for different sports events and proposes a gray-scale modulation–based algorithm to determine lighting demand. In addition, a Neural Network–based control algorithm is developed to enable adaptive and automated lighting adjustments. Experimental results demonstrate that the system effectively optimizes lighting performance across various sporting conditions. The study highlights the potential of intelligent lighting systems to improve energy efficiency and operational effectiveness in sports grounds, indicating strong applicability for real-world implementation. [3]

Rain Detection and Alert Systems

Machine learning models such as LSTM (Long Short-Term Memory) and Bidirectional LSTM are increasingly applied to predict imminent rainfall using historical and real-time weather data, achieving high prediction accuracy, and supporting proactive decision-making. Such predictive capabilities are crucial for cricket pitch curation, where timely rain alerts help protect pitch quality, prevent waterlogging, and optimize maintenance operations. Overall, the literature highlights that integrating Arduino-based sensing with machine learning-driven rain prediction enhances reliability, reduces manual monitoring, and enables intelligent, automated pitch management systems. [4]

Although existing studies demonstrate the effectiveness of individual systems such as soil moisture–based irrigation, intelligent lighting control, and weather monitoring, most of these solutions are developed and implemented as separate, independent systems. This creates limitations in terms of coordination, cost, and overall operational efficiency in real-world environments such as cricket stadiums.

The proposed project addresses this gap by developing a fully integrated environmental control system that combines lighting, irrigation, and rain detection into a single unified platform.

System Methodology

Overview

The system incorporates multiple sensors for real-time data acquisition through a dedicated data acquisition (DAQ) system. The acquired signals are processed and analyzed using LabVIEW software, which facilitates automated decision-making and precise control of the associated actuators.

Sensors Used

Light Sensor (LDR)
- Measures ambient light intensity
- Used to control floodlights automatically

Rain Sensor
- Detects rainfall
- Triggers alert system

Soil Moisture Sensor
- Measures water content in soil
- Controls irrigation system

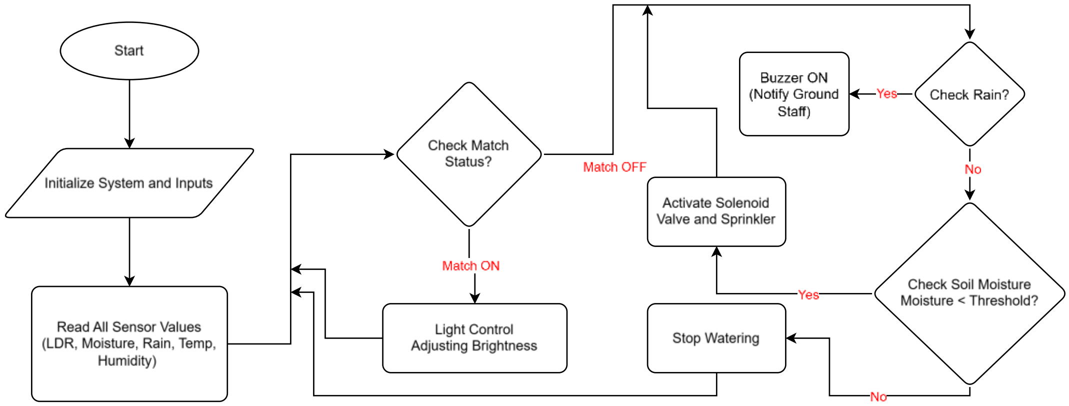
Temperature & Humidity Sensor
- Measures ambient temperature and humidity
- Used to evaluate air moisture and playing conditions

Actuators Used

Linear Servo Actuator
- Automatic up and down motion of water sprinklers

Buzzer
- Provides notification to ground staff in rainy situation

Water Pump
- Automatic control of water flow



Estimated Budget

Component	Quantity	Unit Price (LKR)	Amount (LKR)
LDR 12mm	1	100	100
Rain Sensor Moisture Water Module YL-83 FC-37	1	140	140
Capacitive Soil Moisture Sensor V2.0	2	200	400
HTU21D Temperature and Humidity Sensor Module	1	700	700
Buzzer Module	1	150	150
Water Pump	1	160	160
Optimus Mini Linear Actuator	2	390	780
Toggle Switch	1	100	100
Breadboard & Wires		1000	1000
5V LED Strip	1	800	800
LM3914	4	80	320
Prototyping Accessories		2500	2500
Total Cost (LKR)			7150

Conclusion

This project presents a practical approach to automating key environmental control functions in a cricket stadium. By integrating multiple sensors and actuators with a centralized control system, the proposed solution improves efficiency, reduces manual effort, and enhances safety.
The system demonstrates the application of instrumentation and automation principles in a real-world scenario, making it a valuable learning experience while meeting the project requirements.

References

[1] G. G.L, G. I.E., H. R.L and M. G.L, "Water Application Efficiency and Adequacy of ET-Based and Soil Moisture–Based Irrigation Controllers for Turfgrass Irrigation," *Journal of Irrigation and Drainage Engineering,* vol. 139, no. 2, p. 52, 2012.

[2] R. C.S, R. S and P. J, "IoT-Weather Integration for Enhanced Cricket Tactics with Gradient Boosting Algorithm," in *International Conference on Computational System and Information Technology for Sustainable Solutions*, India, 2024.

[3] Peng DONG, "Application of Intelligent Lighting Control System in Different Sports Events in Sports Venues," *Light and Engineering,* vol. 26, no. 4, pp. 165-171, 2018.

[4] T. S, G. M and N. S, "Weather Monitoring for Curating Cricket Pitches using Machine Learning and Arduino," in *8th International Conference on Computational System and Information Technology for Sustainable Solutions*, Bengaluru, 2024.